File: Dataset S1.xlsx

Sheet: The following variables are provided in the sheet titled "Worker_metabolic_scaling_data" (Includes all data used to generate Figures 1-4, Figure 6 and Supplementary Figures S1-S8 and S10. All units are provided in column headings)

Date: date when worker data were collected

Environment: environment from which worker was collected, including an urban location, field location or colonies reared in-lab

Colony: unique colony identifier

Treatment: whether the worker was sampled following treatment with a food-deprived diet, sampled following treatment with a control diet or sampled within several hours of removal from a queenright colony

HW_millimeter: head width of the worker in units of millimeters

Brain_OCR_measure 1-5: one of 5 replicates of brain oxygen consumption rate recorded for this worker

Brain_OCR_average: average of 5 brain OCR replicates

Brain_Metabolic_Rate_milliliter_per_sec: average brain OCR converted to units of volume

Brain_Metabolic_Rate_microWatts: average brain OCR converted to microWatts and used in analysis

Body_SyringeID: Unique identifier of the chamber containing the worker used for respirometry

CO2_emission_recording_duration_minutes: duration over which the worker was sealed within the respirometry chamber allowing carbon dioxide to accumulate

CO2_emission_flowrate_milliliter_per_min: air flow-rate after chamber containing worker was unsealed, allowing accumulated carbon dioxide to flow through FoxBox CO2 meter for measurement

Integral_milliliterCO2: accumulated CO2 is recorded in parts per million as a signal peak through expedata. The integral of this peak is the associated volume of CO2 emitted by the worker while sealed in the chamber.

Respirometry_injection_volume_milliliter: the volume of air within the respirometry chamber that was injected into the FoxBox following chamber unsealing for measurement

Worker_volume_milliliter: the volume equivalent of the worker's mass

Correctionfactor_for_worker_volume: estimated from worker volume. Because the worker takes up a small amount of the respirometry chamber volume, accurate calculation of the volume of CO2 emitted by the worker needs to be corrected for by this volume, using a factor calculated from the worker's volume. This corrected value is given in the column "Corrected_CO2_emission_milliliter".

Estimated_CO2_leak_milliliter: a standard curve was used to approximate the volume of CO2 leaking over time in each respirometry chamber. This was used to estimate the volume of CO2 that leaked into each chamber that contributed to the total volume of CO2 measured for each worker. This value was subtracted from "Corrected_CO2_emission_milliliter" to determine the actual CO2 emission volume for this worker, given in the column "leak_corrected_CO2_emission_milliliter"

CO2_emission_rate_mlCO2_per_min: CO2 emission volume divided by duration chamber is sealed (Column T) to obtain a rate of CO2 emission.

Calculated_Oxygen_consumption_rate_mlO2_per-min: worker respiratory quotient was used to convert CO2 emission rate to O2 consumption rate for this worker

Worker_metabolic_rate_microWatts: worker body metabolic rate in units of microWatts, used in data analysis

Relative_brain_metabolic_rate: worker brain metabolic rate divided by body metabolic rate (column Q/ column AF)

Relative_brain_mass: worker brain mass divided by body mass (column H/ column J)

Body_mass_specific_metabolic_rate: (column AF/ column I)

Brain_mass_specific_metabolic_rate: (column Q/ column H)

Movement_millimeters: estimated distance (in millimeters) worker moved while sealed in the respirometry chamber

Sheet: The following variables are provided in the sheet titled "behavioral_response_figure_5" (includes all data used to generate Figure 9 and Supplementary Figure S9)

Behavioral_Response_Score: behavioral response score from 0-5 during interaction between *T. immigrans* worker and *P. dentata* minor. “no contact” indicates that there was no interaction and this worker was excluded from analysis of behavioral response.

Sheet: The following variables are provided in the sheet titled "mammal_scaling_data_figure_7" (includes all data used to generate Figure 7)

Brain_mass_grams: brain mass in units of grams

Brain_oxygen_consumption_rate_milliliters_per_min: brain OCR in units of milliliters per minute. This is the mean value for colony standards estimated at 37 degrees Celsius assuming a Q_{10} of 2.

Sheet: The following variables are provided in the sheet titled "Mortality_data_figS11" (includes all data used to generate Supplementary Figure S11)

Start Date: indicates the date on which a treatment for a Subcolony was started

Subcolony: individual identifier of the group of 30 workers exposed to either a control or food deprivation diet

Day and Status: indicates the number of workers dead on a given number of days following the start date. “1” indicates a worker is dead. “0” indicates no worker is dead.

The data can be interpreted with the following examples

: Subcolony 7 was provided a food-deprived treatment beginning on Jun 24 2018. On day 1 of the treatment, one worker was found dead and therefore one row (row 2) has a status of 1 (cell F2) for day 1. On day 7, 6 workers were found dead and therefore 6 rows (rows 8-13) have a status of 1 (cells F8-13) for day 7. By day 10, a total of 26 workers were dead and therefore Subcolony 7 has a total of 26 rows. Because Subcolony 7 was initially provided with 30 workers, the ending mortality of this subcolony was 26/30 = 87%.

: Subcolony C1 was provided a control diet beginning on Feb 6 2024 (beginning on row 3293). No worker died until day 16, the end of the treatment, and on day 16 only one worker was dead. Therefore, Subcolony C1 has one row (row 3293) with a status of 1 (cell F3293) and 29 rows with a status of 0 (cells 3294-3322). The ending mortality on day 16 for Subcolony C1 was 1/30 = 3%.

Sheet: The following variables are provided in the sheet titled "Stable_brain_traces_figS12" (includes all data used to generate Figure S12)

OCR_Time_Point: Minute (in decimal) during which a worker's brain oxygen consumption rate was recorded

Colony Standard_1-5, Food_Deprived_1-5: Brain oxygen consumption rates of each of 5 workers for each Treatment recorded at each time point. Note that Colony Standard_4 and Colony Standard_5 were recorded at different time intervals from the other workers and their brain OCR values are associated with different time points (column K as opposed to column A).